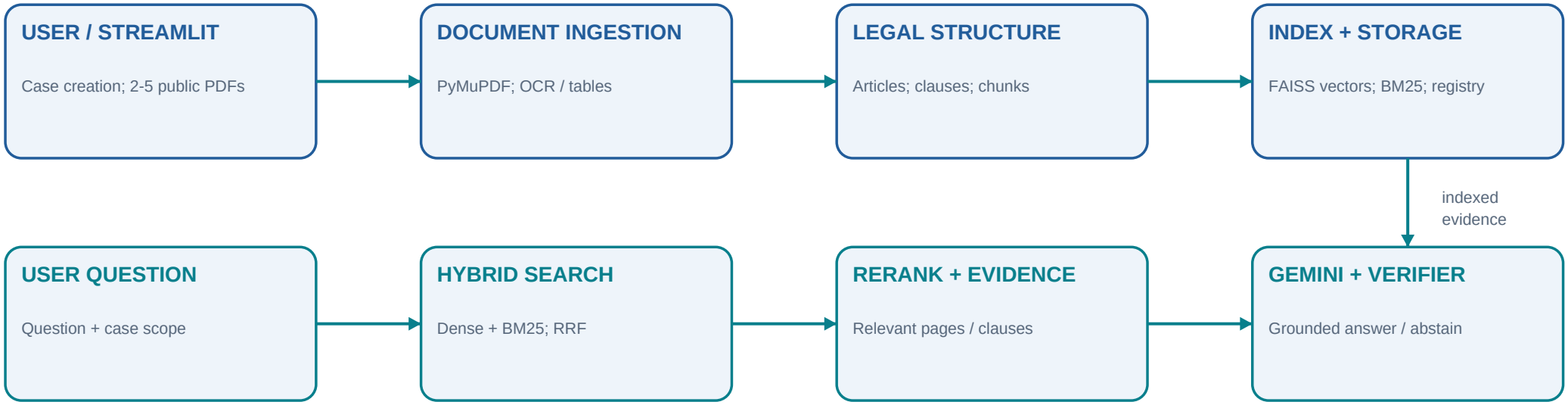
COMPLETE SYSTEM ARCHITECTURE

End-to-end legal document intelligence | two linked pipelines: indexing and question answering



WHAT HAPPENS

The top row prepares documents once. The bottom row answers each question using indexed evidence.

EXAMPLE

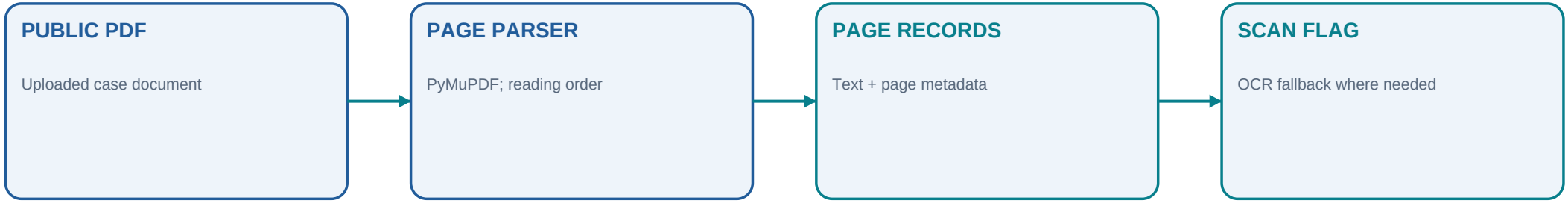
Upload GDPR -> extract Article 83 -> create searchable chunks -> ask about fine tiers -> retrieve both clauses -> cite pages.

WHAT TO SAY

Our architecture separates document preparation from answer generation, and verifies citations before presenting results.

PHASE 1 - PDF INGESTION & EXTRACTION

Turn public legal PDFs into readable page records



WHAT HAPPENS

The system reads each page, preserves its page number and identifies pages with little extractable text.

EXAMPLE

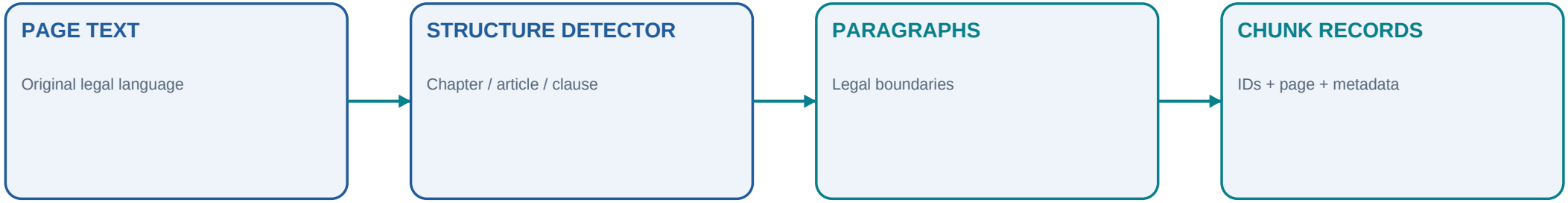
An 88-page GDPR PDF becomes 88 page records. An image-only page is flagged for OCR.

WHAT TO SAY

PDF ingestion is not just copying text: we preserve page-level provenance so later citations can be checked.

PHASE 2 - LEGAL STRUCTURE & CHUNKING

Preserve the hierarchy lawyers use



WHAT HAPPENS

The pipeline groups text by legal sections rather than splitting solely by a fixed character count.

EXAMPLE

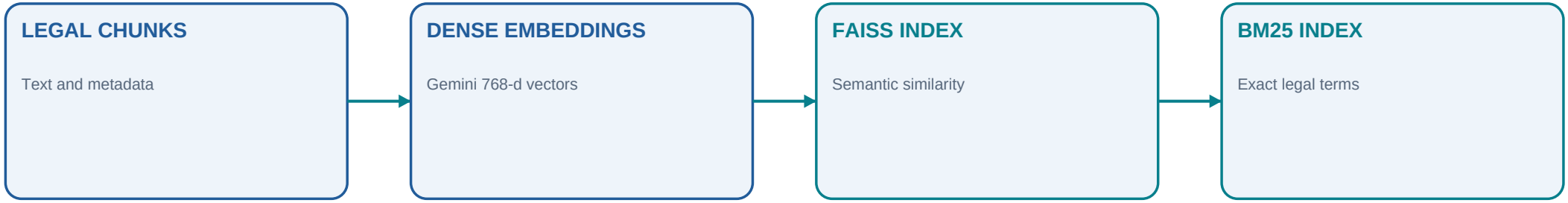
Article 83(4) and Article 83(5) remain individually traceable for a two-tier GDPR fine question.

WHAT TO SAY

Clause-aware chunking makes references like Section 7.2 easier to retrieve and cite.

PHASE 3 - EMBEDDINGS & INDEXING

Build two complementary search paths



WHAT HAPPENS

Dense vectors capture meaning; BM25 captures literal clause numbers and legal keywords.

EXAMPLE

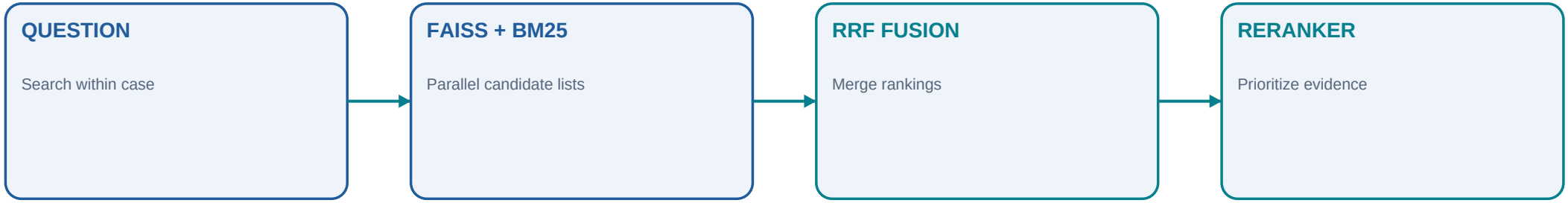
A question about penalties can find semantically similar text, while 'Article 83(5)' is an exact-match signal.

WHAT TO SAY

We combine semantic understanding with lexical precision rather than trusting only one retrieval method.

PHASE 4 - HYBRID RETRIEVAL & RERANKING

Find the most useful evidence for a question



WHAT HAPPENS

FAISS and BM25 return candidates. Reciprocal Rank Fusion merges rankings; optional cross-encoder reranking refines order.

EXAMPLE

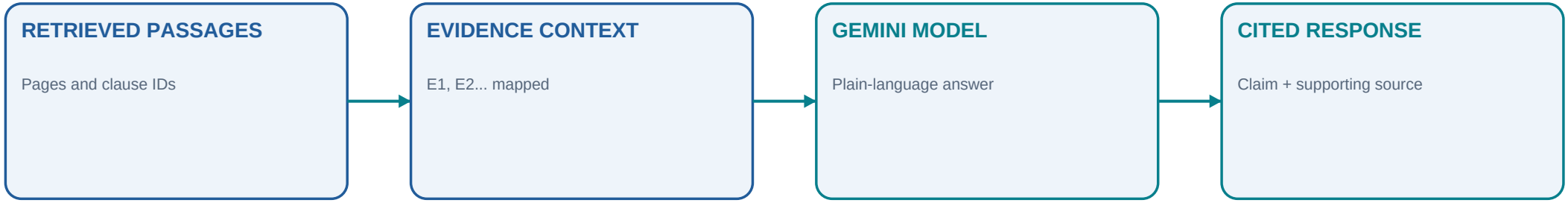
For GDPR fine tiers, separate candidates for Article 83(4) and 83(5) can both enter the evidence set.

WHAT TO SAY

Our provisional GDPR benchmark found hybrid Hit@5 of 100% versus BM25 Hit@5 of 75%.

PHASE 5 - GROUNDED ANSWER GENERATION

Answer from retrieved documents, not memory



WHAT HAPPENS

The generator receives a constrained evidence package with source metadata and produces a cited answer.

EXAMPLE

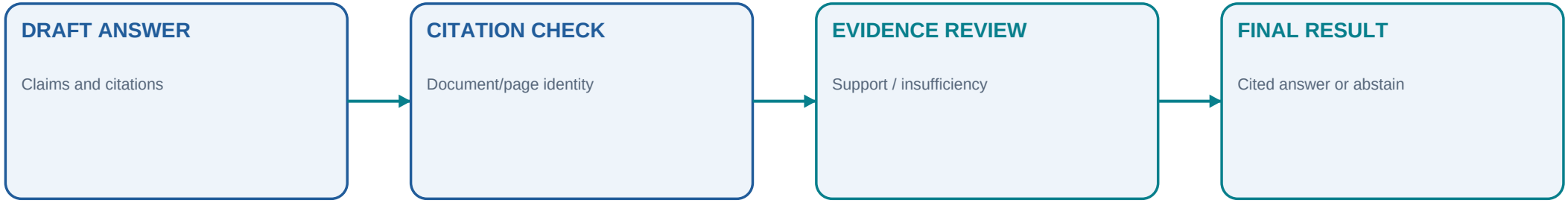
The response explains the two GDPR fine tiers using the passages for Article 83(4) and (5).

WHAT TO SAY

The model is instructed to use only retrieved evidence, and citations remain tied to their source metadata.

PHASE 6 - CITATION VERIFICATION & ABSTENTION

Check support and avoid unsupported claims



WHAT HAPPENS

Citation checks ensure referenced evidence belongs to the correct document; insufficient evidence can trigger refusal.

EXAMPLE

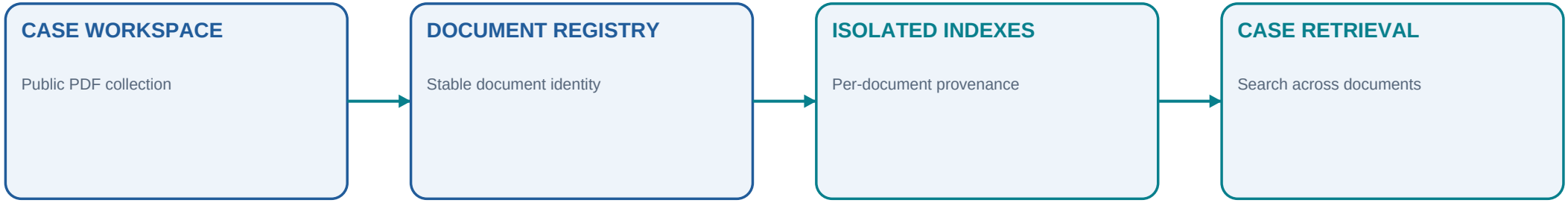
Ask for the governing law of a subcontractor agreement not uploaded: the safe response is insufficient evidence.

WHAT TO SAY

We implemented verification and abstention handling; their accuracy has not yet been formally measured.

PHASE 7 - MULTI-DOCUMENT CASE MANAGEMENT

Keep different PDFs and their evidence separate



WHAT HAPPENS

The case layer associates each document with its own identity so identical filenames do not merge evidence.

EXAMPLE

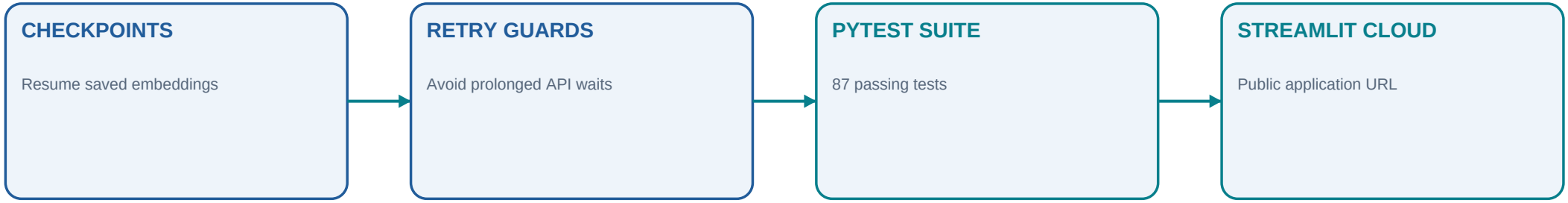
Two different files both named Contract.pdf still retain different document IDs and citations.

WHAT TO SAY

Document identity propagation was regression-tested while preserving the original GDPR 397-chunk baseline.

PHASE 8 - RESILIENCE, TESTS & DEPLOYMENT

Make the prototype repeatable and demonstrable



WHAT HAPPENS

Checkpointing and retry safeguards improve recovery; automated tests protect the established behavior.

EXAMPLE

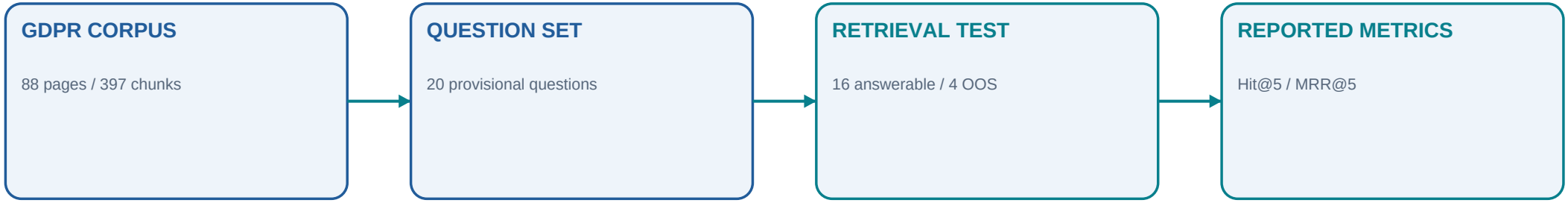
A failed embedding run can use saved work; 87 tests passed in the final local validation.

WHAT TO SAY

The public demo exists, but new PDF processing can be limited by Gemini quota and Streamlit's ephemeral storage.

PHASE 9 - EVALUATION & EVIDENCE

Measure retrieval and disclose what remains untested



WHAT HAPPENS

We measured retrieval against a small provisional benchmark, not full legal-answer accuracy.

EXAMPLE

BM25: Hit@5 75%, MRR@5 .5073; hybrid: Hit@5 100%, MRR@5 .7917.

WHAT TO SAY

Next evaluation priorities are held-out multi-hop questions, answer correctness, citation precision, abstention accuracy and latency.